A vortex obstruction to openness of the wave-function–density map

A counterexample to the unqualified question in arXiv:1907.02024

Autonomous mathematical discovery run `fa_banach_001`, lane 5

12 August 2026

Status: candidate counterexample, likely valid. The construction is exact and analytic. It settles the question as stated without an $N \geq 2$ restriction, already for one spinless particle.

1 Source question

In *From wave-functions to single electron densities*, Ugo Bindini and Luigi De Pascale consider the maps from symmetric and antisymmetric normalized H^1 wave functions to their one-particle densities [1, p. 2]. They ask:

> Suppose that $\sqrt{\rho_k} \to \sqrt{\rho}$ in H^1, and take ψ such that $\rho = \rho[\psi]$. Can one find ψ_k such that $\rho_k = \rho[\psi_k]$ and $\psi_k \to \psi$ in H^1? In other words, are $\Phi^{\mathcal{A}}$ and $\Phi^{\mathcal{S}}$ open?

Their Theorems 1.1–1.2 give positive answers for nonnegative and real-valued bosonic wave functions. The 2026 journal version, retitled *From Wave-Functions to Single Boson Densities*, still explicitly describes the general question as open [2]. Figure 1 shows the source formulation and its partial results.

2 Proof intuition

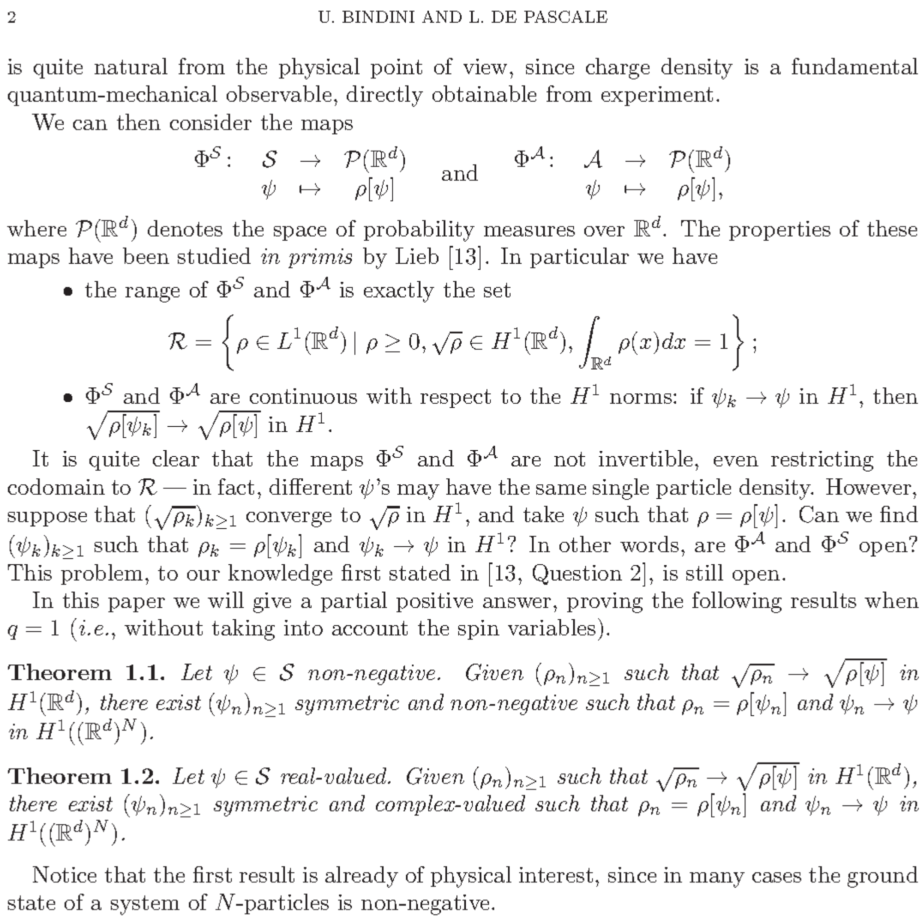
For one spinless particle, the density is simply $\rho[\psi] = |\psi|^2$. A complex wave function can carry phase topology invisible to its density. Take a compactly supported wave function whose phase winds once around the x_1x_2-axis. Its modulus vanishes on that axis. Fill the zero with an arbitrarily small positive core; the new square-root density converges in H^1, but a wave function having that positive modulus has an H^1 circle-valued phase across every transverse disk. Such a disk phase has degree zero on interior circles. It therefore cannot converge in H^1 to the degree-one vortex phase.

The point is specific and sharp: the source's positive theorem covers real phases, while the obstruction is genuinely complex. It is also invisible at the density level.

3 A Sobolev phase lemma

Lemma 1 (No H^1 vortex without a zero)**.** *Let $D \subset \mathbb{R}^2$ be a disk and $v \in H^1(D; \mathbb{S}^1)$. Then $v = e^{i\varphi}$ for some $\varphi \in H^1(D; \mathbb{R})$. Consequently, for almost every concentric circle $\partial D_r \subset D$, the restriction of v has degree zero.*

Proof. Set $j = -i\overline{v}\,\nabla v \in L^2(D; \mathbb{R}^2)$. Differentiating $|v|^2 = 1$ shows that each weak derivative of v is tangent to $\mathbb{S}^1$, so $\partial_k v = i j_k v$ almost everywhere. The distributional curl of j is zero: after



2 U. BINDINI AND L. DE PASCALE

is quite natural from the physical point of view, since charge density is a fundamental quantum-mechanical observable, directly obtainable from experiment.

We can then consider the maps

$$\begin{array}{rccc} \Phi^{\mathcal{S}}: & \mathcal{S} & \to & \mathcal{P}(\mathbb{R}^d) \\ & \psi & \mapsto & \rho[\psi] \end{array} \quad \text{and} \quad \begin{array}{rccc} \Phi^{\mathcal{A}}: & \mathcal{A} & \to & \mathcal{P}(\mathbb{R}^d) \\ & \psi & \mapsto & \rho[\psi], \end{array}$$

where $\mathcal{P}(\mathbb{R}^d)$ denotes the space of probability measures over $\mathbb{R}^d$. The properties of these maps have been studied *in primis* by Lieb [13]. In particular we have

- the range of $\Phi^{\mathcal{S}}$ and $\Phi^{\mathcal{A}}$ is exactly the set

$$\mathcal{R} = \left\{ \rho \in L^1(\mathbb{R}^d) \mid \rho \geq 0, \sqrt{\rho} \in H^1(\mathbb{R}^d), \int_{\mathbb{R}^d} \rho(x)dx = 1 \right\};$$

- $\Phi^{\mathcal{S}}$ and $\Phi^{\mathcal{A}}$ are continuous with respect to the H^1 norms: if $\psi_k \to \psi$ in H^1, then $\sqrt{\rho[\psi_k]} \to \sqrt{\rho[\psi]}$ in H^1.

It is quite clear that the maps $\Phi^{\mathcal{S}}$ and $\Phi^{\mathcal{A}}$ are not invertible, even restricting the codomain to $\mathcal{R}$ — in fact, different ψ's may have the same single particle density. However, suppose that $(\sqrt{\rho_k})_{k\geq 1}$ converge to $\sqrt{\rho}$ in H^1, and take ψ such that $\rho = \rho[\psi]$. Can we find $(\psi_k)_{k\geq 1}$ such that $\rho_k = \rho[\psi_k]$ and $\psi_k \to \psi$ in H^1? In other words, are $\Phi^{\mathcal{A}}$ and $\Phi^{\mathcal{S}}$ open? This problem, to our knowledge first stated in [13, Question 2], is still open.

In this paper we will give a partial positive answer, proving the following results when $q = 1$ (*i.e.*, without taking into account the spin variables).

Theorem 1.1. *Let $\psi \in \mathcal{S}$ non-negative. Given $(\rho_n)_{n\geq 1}$ such that $\sqrt{\rho_n} \to \sqrt{\rho[\psi]}$ in $H^1(\mathbb{R}^d)$, there exist $(\psi_n)_{n\geq 1}$ symmetric and non-negative such that $\rho_n = \rho[\psi_n]$ and $\psi_n \to \psi$ in $H^1((\mathbb{R}^d)^N)$.*

Theorem 1.2. *Let $\psi \in \mathcal{S}$ real-valued. Given $(\rho_n)_{n\geq 1}$ such that $\sqrt{\rho_n} \to \sqrt{\rho[\psi]}$ in $H^1(\mathbb{R}^d)$, there exist $(\psi_n)_{n\geq 1}$ symmetric and complex-valued such that $\rho_n = \rho[\psi_n]$ and $\psi_n \to \psi$ in $H^1((\mathbb{R}^d)^N)$.*

Notice that the first result is already of physical interest, since in many cases the ground state of a system of N-particles is non-negative.

Figure 1: The open question and partial positive results on p. 2 of the source paper.

cancellation of mixed derivatives, the remaining expression is

$$-i(\partial_1\overline{v}\,\partial_2 v - \partial_2\overline{v}\,\partial_1 v) = 0.$$

The Sobolev Poincaré lemma on the simply connected disk gives $\varphi \in H^1(D)$ with $\nabla\varphi = j$. Hence $\nabla(e^{-i\varphi}v) = 0$, and changing φ by a constant yields $v = e^{i\varphi}$.

For almost every r, the circular trace $\varphi(r,\cdot)$ belongs to $H^1(\mathbb{S}^1)$ and is periodic. Therefore

$$\deg(v|_{\partial D_r}) = \frac{1}{2\pi i}\int_0^{2\pi} \overline{v}\,\partial_\theta v\,d\theta = \frac{1}{2\pi}\int_0^{2\pi} \partial_\theta\varphi\,d\theta = 0.$$

$\square$

4 Counterexample

Theorem 1. *In the spinless one-particle sector, for every spatial dimension $d \geq 2$, neither the symmetric nor the antisymmetric wave-function–density map is open in the topology used in [1]. More precisely, there are a normalized $\psi \in H^1(\mathbb{R}^d;\mathbb{C})$ and probability densities ρ_n such that*

$$\sqrt{\rho_n} \longrightarrow \sqrt{\rho[\psi]} \quad \text{in } H^1(\mathbb{R}^d),$$

but no choice of $\psi_n \in H^1(\mathbb{R}^d;\mathbb{C})$ satisfying $\rho[\psi_n] = \rho_n$ can converge to ψ in H^1.

Proof. Write $x = (z, y)$ with $z = (x_1, x_2) \in \mathbb{R}^2$ and $y \in \mathbb{R}^{d-2}$ (omit y when $d = 2$). Choose nonnegative smooth compactly supported cutoffs $\chi(z)$ and $\zeta(y)$ which equal one on $|z| \le 3$ and on a nonempty cube $Q \subset \mathbb{R}^{d-2}$, respectively, with χ radial. For $d = 2$, set $\zeta \equiv 1$. Let

$$\psi(z, y) = C_0\,(x_1 + ix_2)\chi(z)\zeta(y),$$

where $C_0 > 0$ normalizes the L^2 norm to one. Its modulus is

$$a(z, y) := |\psi(z, y)| = C_0|z|\chi(z)\zeta(y).$$

For $\varepsilon > 0$, define

$$a_\varepsilon(z, y) = C_\varepsilon\sqrt{|z|^2 + \varepsilon^2}\,\chi(z)\zeta(y), \qquad \rho_\varepsilon = a_\varepsilon^2,$$

where C_ε makes $\|a_\varepsilon\|_2 = 1$. Plainly $C_\varepsilon \to C_0$. Moreover,

$$\nabla_z\sqrt{|z|^2 + \varepsilon^2} = \frac{z}{\sqrt{|z|^2 + \varepsilon^2}} \longrightarrow \frac{z}{|z|} \quad \text{a.e.},$$

and these gradients are bounded by one. Applying dominated convergence also to the cutoff terms gives

$$a_\varepsilon \longrightarrow a = |\psi| \quad \text{strongly in } H^1(\mathbb{R}^d). \tag{1}$$

Thus ρ_ε belongs to the source's admissible density class and has the required convergence.

Take any sequence $\varepsilon_n \downarrow 0$ and suppose, for contradiction, that $\psi_n \to \psi$ in H^1 while $|\psi_n| = a_{\varepsilon_n}$ almost everywhere. On the cylinder

$$\Omega = \{\, |z| < 3 \,\} \times Q$$

the modulus a_{ε_n} is strictly positive. Hence

$$u_n := \frac{\psi_n}{a_{\varepsilon_n}} \in H^1(\Omega; \mathbb{S}^1).$$

On the annular cylinder

$$\Omega' = \{\, 1 < |z| < 2 \,\} \times Q,$$

the functions a_{ε_n} and their reciprocals converge in $W^{1,\infty}$ to $C_0|z|$ and $(C_0|z|)^{-1}$. It follows from $\psi_n \to \psi$ that

$$u_n \longrightarrow u(z, y) := \frac{x_1 + ix_2}{|z|} \quad \text{strongly in } H^1(\Omega'). \tag{2}$$

Pass to a subsequence for which the squared $H^1(\Omega')$ errors in (2) are summable. Fubini's theorem first in y and then in the polar radius produces one $y_0 \in Q$ (unneeded for $d = 2$) and one $r_0 \in (1, 2)$ such that

$$u_n(r_0, \cdot, y_0) \longrightarrow e^{i\theta} \quad \text{strongly in } H^1(\mathbb{S}^1),$$

and such that every disk slice $u_n(\cdot, y_0)$ lies in $H^1(\{|z| < 3\}; \mathbb{S}^1)$. Lemma 1 gives

$$\deg u_n(r_0, \cdot, y_0) = 0 \quad \text{for every } n.$$

But degree is continuous under strong $H^1(\mathbb{S}^1)$ convergence: the formula

$$\deg w = \frac{1}{2\pi i}\int_0^{2\pi} \overline{w}\, w'\, d\theta$$

passes to the limit because $H^1(\mathbb{S}^1)$ embeds into $C^0(\mathbb{S}^1)$. The limit $e^{i\theta}$ has degree one, a contradiction.

For $N = 1$ the permutation group is trivial, so the same ψ belongs to both the symmetric and antisymmetric classes. Finally, failure of the stated sequential lifting property implies non-openness: if either map were open at ψ, a diagonal choice from preimages of the approaching ρ_{ε_n} would produce the forbidden convergent lifting sequence. □

5 Verification, novelty, and scope

Every step above is qualitative and exact; no numerical verification is needed. The two points most worth checking are the H^1 lifting lemma and the Fubini selection of a common two-dimensional slice and circle. The latter is standard: summability of the annular H^1 errors makes the sliced errors tend to zero for almost every (y, r), while the trace and lifting conclusions fail only on countable unions of null sets.

A bounded novelty search on 12 August 2026 covered the run's registry, solution, attempt, and proof-gap indexes; arXiv and current web searches for the exact title and question; and combinations of "Sobolev modulus map," "H^1 vortex degree," "Lieb Question 2," and "counterexample." No prior counterexample was located. The May 2026 journal article still calls the general question open and says that its vectorial/phase step is incomplete [2]. This supports, but cannot guarantee, novelty.

Scope limitation. The theorem disproves the question as formulated without excluding one-particle systems. It does *not* settle a reformulation imposing $N \geq 2$. In a many-particle disintegration, the normalized conditional wave function takes values in an infinite-dimensional unit sphere, so the scalar $\mathbb{S}^1$ degree obstruction used here need not persist. Human review should first confirm that $N = 1$ lies within the intended quantifiers of Lieb's and the source paper's question. If an implicit many-particle convention excludes it, this remains a sharp one-particle obstruction rather than a full answer to that restricted variant.

References

[1] U. Bindini and L. De Pascale, *From wave-functions to single electron densities*, arXiv:1907.02024, 2019. https://arxiv.org/abs/1907.02024.

[2] U. Bindini and L. De Pascale, *From Wave-Functions to Single Boson Densities*, Archive for Rational Mechanics and Analysis 250 (2026), Article 44. https://doi.org/10.1007/s00205-026-02207-2.

[3] E. H. Lieb, *Density functionals for Coulomb systems*, International Journal of Quantum Chemistry 24 (1983), 243–277.